ÜSKÜDAR AMERICAN ACADEMY
GRADE 10 CHEMISTRY WORKSHEET # 2

SUBJECT: ***MOLE CONCEPT***

Express your answers in scientific notation and in three significant figures.

1. A C_2X_4 gas sample that occupies a volume of 5.60 liters at STP, weighs 25 grams.

   a) What is molar mass of C_2X_4? (Ans: 100 g/mol)
   b) What is the molar mass of X? (Ans: 19.0 g/mol)

   $\frac{5.6}{22.4} = 0,25$ mol　　$0,25 = \frac{25}{MM}$
   $12 \times 2 = 24$　　$0.25 MM = 25$
   $100 - 24 = 76$　　$MM = 100$
   $76/4 = 19$

2. 0.32 moles of X_2O_5 compound weigh 34.57 grams. Determine the atomic mass of X. (Ans: 14.0 g/mol)

   $\frac{34.57}{0.32} = 108$　　$2x + 80 = 108$
   $2x = 28$
   $x = 14$ g/mol

3. 6.95 grams of X_3O_4 compound contains 1.26×10^{23} atoms. What is the molar mass of X? (Ans: 55.9 g/mol)

   mol $= \frac{1.26 \times 10^{23}/7}{6.02 \times 10^{23}} = 0,03$　　$0,03 = \frac{6.95}{MM} \rightarrow MM = 231,6$
   $231,6 - 3(16) = 55,6 \frac{g}{mol}$

4. How many He atoms occupy 14.56 liters at STP? (Ans: 3.91×10^{23})

   $\frac{14.56}{22.4} = 0.65 \rightarrow 0.65 \times 6,02 \times 10^{23} = 3,91 \times 10^{23}$

5. What is the volume of 6.02×10^{21} molecules of NO_2 gas at STP? (Ans: 0.224 L)

   $\frac{6.02 \times 10^{21}}{6.02 \times 10^{23}} = \frac{vol}{22.4} \rightarrow 100$ vol $= 22.4$
   100　　volume $= 0.224$ L

6. 5.40 g of A_2O_5 gas occupies 1.12 L at STP. Calculate the atomic mass of A. (Ans: 14.0 g/mol)

   $\frac{5.4}{MM} = \frac{1.12}{22.4} \rightarrow 1.12 MM = 120,96$　　$16.5 = 80$
   $MM = 108$　　$108 - 80 = 28$
   $28/2 = 14$ g/mol

7. 0.20 mol of AO and 0.30 mol AO_2 has a total mass of 18.80 g. What is the atomic mass of element A? (Ans: 12 g/mol)

   $0.30 \times 2 = 0,60$
   $+ 0,20$
   $0,80 = \frac{m}{16} = 12,8$ g
   $18,8 - 12,8 = 6$ g
   $0,30 + 0,20 = 0,50$　　$0,50 = \frac{6}{MM} = 0,5 MM = 6 \rightarrow$ molar mass $= 12$ g/mol

8. How many oxygen atoms are present in a mixture composed of 0.40 mol N_2O_3, 16 g SO_3, and 5.60 L of CO_2 at STP? (Ans: 1.38×10^{24})

   $0,40 \times 3 = 1,20$ mol　　$\frac{5.60}{22.4} = 0.25$　　$0,50$
   $\frac{16}{80} = 0,20 \rightarrow 0,20 \times 3 = 0,60$ mol　　$0,25 \times 2 = 0,50$ mol　　$1,20$
   $+ 0,60$
   2.30 mol $\times 6.02 \times 10^{23} = 1,384 \times 10^{24}$

9. 20 g of $XO_{3\,(g)}$ occupies a volume of 5.60 L at STP. Find the atomic mass of X. (Ans: 32.0 g/mol)

   $\frac{5.60}{22.4} = 0,25$ mol　　$0,25 \times 3 = 0,75$ mol
   $0,75 \times 16 = 12$ g
   20g $- 12$g $= 8$ gram X $\rightarrow 8.4 = 32$ g/mol

10. 36 % by mass of Al_2X_3 is Al metal. What is the molar mass of X element? (Ans: 32.0 g/mol)

   54
   54 36　　$\frac{54 \cdot 64}{36} = 32$　　$\frac{1800}{\cancel{5400}}$ $\frac{1}{\cancel{3}}$
   64　　36

11. 0.5 mol of X atoms combine with 12.0 g of O atoms in X_2O_n. What is the value of "n"? (Ans: n=3)

   $12 = 0.75$　　$\frac{12}{16} = 0,75$
   $X \rightarrow 0,5$　　2
   $O \rightarrow 0,75$　　3

12. An organic compound composed of C, H and O atoms weighs 0.92 grams and contains 0.24 grams C and 24.10^{21} H atoms. If the given mass of the compound contains $6x10^{21}$ molecules, what is the molecular formula of the compound? (Ans: $C_2H_4O_4$)

13. 4.92 grams of the hydrate; $MgSO_4.xH_2O$; is heated to lose its water and weighed again as 2.40 g. What is the value of "x" in the formula? (Ans: x=7)

14. The analysis of a compound showed that it was made up of 7.2 g C, 1.2 mol of H, and 3.63 x 10^{23} atoms of oxygen.

a) What is the **molecular formula** of the compound if its molar mass is 180 g/mol? (Ans: $C_6H_{12}O_6$)
b) What is the percent composition by mass of this compound? (Ans: 40.0 % C, 6.70 % H, 53.3 % O)

$\frac{37}{12} = 3,0$ $\quad \frac{2.20}{1} = 2,2$ $\quad \frac{18,5}{14} = 1,32$ $\quad \frac{42.3}{16} = 2,6$

15. The explosive, TNT, is composed of 37.0 % carbon, 2.20 % hydrogen, 18.5 % nitrogen and 42.3 % oxygen by mass.

a) Determine the **empirical formula** for TNT. (Ans: $C_7H_5N_3O_6$)

b) The molar mass for TNT is 227 g/mol. What is the **molecular formula** for TNT? (Ans: $C_7H_5N_3O_6$)

$\frac{3,0}{1,32}$ $\quad \frac{2,2}{1,32}$ $\quad \frac{1,32}{1.32}$ $\quad \frac{2,6}{1.32}$ $\quad C_7H_5N_3O_6$

$\frac{7}{3}$ $\quad \frac{5}{3}$ $\quad 1$ $\quad 2$

$\times 3$

$= 7 \quad 5 \quad 3 \quad 6$